SATYANAND SHUKLA

AI/ML Engineer | Generative AI, Agentic RAG & LLM Systems | Computational Genomics

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PROFESSIONAL SUMMARY

AI/ML Engineer specializing in **Agentic RAG, LLM fine-tuning, and self-corrective retrieval systems** — building production-grade multi-agent platforms with **Self-RAG, Corrective-RAG, prompt engineering, and RAG evaluation (RAGAS, DeepEval)** to ground LLM outputs and minimise hallucination. Experienced with **multimodal LLMs** and real-time **voice/audio AI** (streaming ASR, VAD, TTS), alongside **multi-agent orchestration (LangChain, LangGraph, MCP)** and scalable FastAPI microservice deployment on **AWS/cloud infrastructure**. Applies this generative-AI stack to computational genomics and biomedical data, with an M.Tech in Bioinformatics (MANIT Bhopal, CGPA 9.2/10) and a B.Tech in Computer Science, and a proven record of shipping efficient ML/DL pipelines across genomic, clinical, and imaging datasets.

EXPERIENCE

Software Engineer — AIGENOMICS INDIA PRIVATE LIMITED, Pune Jul 2026 – Present

- Built 40+ FastAPI microservices for AI-driven genomic pipelines using Python, SQLAlchemy QueuePool, and Redis caching.
- Architected stateless JWT OAuth2 authentication with RBAC permissions and dual-layer AES-256-GCM and Fernet PII encryption.
- Engineered POSIX file-locked background schedulers (fcntl.flock) for automated AWS S3 data sync, PDF report generation, and invoicing.
- Deployed containerized microservices using Docker on Linux with custom ASGI API audit logging middleware for seamless Angular integration.

AI/ML Engineer Intern — Get Your Lawyers, Bhopal Apr 2026 – Jun 2026

- Built an **Agentic RAG-based Legal AI platform** using fine-tuned LLMs, Qdrant vector database, **Self-RAG and Corrective-RAG** for Indian legal question answering, with claim verification and bounded re-retrieval to reduce unsupported answers.
- Applied **prompt engineering** techniques — instruction design, few-shot templates, and structured-output prompting — to optimise LLM behaviour for query understanding, legal reasoning, and answer generation across diverse case types.
- **Fine-tuned domain-specific LLMs** on Indian legal data using LoRA/PEFT; implemented retrieval-grounded generation to minimise hallucinations and improve factual accuracy.
- Designed automated ingestion pipelines for PDF parsing, chunking, embedding generation, metadata extraction, and vector indexing across large-scale legal corpora.
- Engineered **multi-agent workflows** using LangChain and LangGraph, covering query understanding, document retrieval, answer verification, and legal reasoning; improved response quality through retrieval validation, context-aware reranking, and corrective feedback mechanisms.

EDUCATION

M.Tech, Bioinformatics — Maulana Azad National Institute of Technology (MANIT), Bhopal CGPA: 9.2/10 | 2024 – 2026

B.Tech, Computer Science — People's University, Bhopal CGPA: 7.91/10 | 2024

TECHNICAL SKILLS

Programming: Python, JavaScript, TypeScript
Deep Learning: PyTorch, TensorFlow, Keras, OpenCV, PyTorch Geometric, CNN, RNN, LSTM, ANN, GNN, Transfer Learning
NLP & Generative AI: NLP, Transformers (BERT), Hugging Face, LLMs, Multimodal LLMs, Multi-LLM Orchestration, LLM Fine-tuning, LoRA/PEFT, Prompt Engineering, LLM Evaluation
RAG & Retrieval: RAG, Agentic RAG, Hybrid RAG, Self-RAG, Corrective-RAG, Embeddings, Search Algorithms, Query Routing, Reranking, RAG Evaluation
Agentic AI: AI Agents, Multi-Agent Systems, CrewAI, AutoGen, Semantic Kernel, LangChain, LangGraph, MCP (Model Context Protocol), Tool/Function Calling, Agent Workflows, Guardrails
Voice / Audio AI: Streaming ASR (faster-whisper), Silero VAD, WebRTC, Streaming TTS, Barge-in Handling, TTFA Optimisation
Web & APIs: FastAPI, REST APIs, WebSockets, Microservices, Postman, React.js, Next.js, Node.js
Databases & Vector DB: MySQL, MongoDB, Qdrant, FAISS, Pinecone, Chroma
Cloud, DevOps & Infra: AWS (S3, EC2), Azure, GCP, Git, GitHub, Docker, CI/CD, Scalable Deployment Pipelines
LLM Ops & Evaluation: RAGAS, DeepEval, Promptfoo, OpenAI API, Gemini API, MLflow, Weights & Biases (W&B), Experiment Tracking, Model Monitoring
Genomics & Drug Discovery: Genome Assembly, Variant Calling, Sequence Alignment, SNP, Molecular Docking, Molecular Dynamics (MD) Simulation, Biomathematics
Machine Learning: Supervised/Unsupervised Learning, Feature Engineering, Model Evaluation, Ensemble Learning, scikit-learn, XGBoost, LightGBM, Bagging/Boosting, Underfitting-Overfitting, Efficient ML/DL Pipelines
Mathematics: Statistics, Probability, Linear Algebra, Calculus, Discrete Mathematics
Data & Visualisation: Pandas, NumPy, ETL, Data Pipelines, Data Wrangling, Matplotlib, Seaborn, Plotly, EDA, Feature Engineering
CSE Fundamentals: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, OOP

PROJECTS

Multimodal Self-Corrective Agentic RAG for Genomic LIMS Intelligence
Python · LangGraph · FastAPI · Qdrant · Elasticsearch · Neo4j · LoRA/QLoRA · Multimodal LLM · RAGAS · DeepEval · LangSmith

- Built a LIMS-integrated multimodal Agentic RAG system combining structured genomic test results with curated scientific knowledge, using hybrid BM25 + dense retrieval, GraphRAG, cross-encoder reranking, and context compression to ground genomic-result explanations in evidence.
- Implemented **self-corrective RAG** with query reformulation, evidence-sufficiency checks, claim/citation verification, and bounded re-retrieval/abstention; added **multi-agent orchestration** (Planner, Research, Verification agents) and **LoRA/QLoRA fine-tuning** with adaptive model routing to reduce hallucination and unnecessary LLM/tool calls.
- Developed an evaluation and observability framework using RAGAS, DeepEval, and LangSmith/OpenTelemetry, measuring retrieval quality, groundedness, hallucination rate, agent task success, tool efficiency, P95 latency, token usage, and cost.

AI-Driven Genomic Drug Response Prediction Platform
Python · Streamlit · Pandas · Rule Engine · OCR

- Building a clinical bioinformatics web platform that predicts personalised drug response from patient biomarkers using a 56-module structured biological knowledge base spanning 12 genomic domains.
- Engineered a dynamic multi-tier rule engine: biological rules → mechanism IDs → clinical mappings → narrative interpretations; specificity-first condition matching with compound AND/OR logic.
- Integrated OCR-based lab report parsing (pytesseract) for automated biomarker extraction and form auto-fill; designed a premium multi-page interface with summary tables, AI assistant, and multi-format export (HTML, CSV, JSON).

SGHD: GNN + Deep Learning for Gut Microbiome-Based ASD Prediction
Python · PyTorch · XGBoost · LightGBM · SHAP

- Built an end-to-end metagenomics pipeline for microbiome biomarker discovery and ASD prediction using Deep Autoencoders, Graph Neural Networks (GNN), and ensemble learning.
- Identified 57 statistically significant microbial biomarkers (FDR < 0.05); validated community differences via PERMANOVA on 16S rRNA/WGS data; achieved ROC-AUC 0.922 (95% CI: 0.811–0.992) on an independent test set, with SHAP for biological interpretation.

Brain Tumour MRI Classification — Hybrid CNN–GNN–SVM (M.Tech Dissertation)
Python · PyTorch · PyTorch Geometric · scikit-learn

- Developed HybridDEGS, integrating DenseNet121, Graph Attention Network (GAT), and calibrated RBF-SVM for automated brain tumour classification from MRI images.
- Designed a preprocessing pipeline with percentile normalization, CLAHE, bilateral filtering, Sobel–Canny edge fusion (4-channel input), duplicate removal (SHA-1, pHash, FAISS), and class-balanced augmentation to prevent data leakage.
- Constructed spatial + feature-similarity graphs, trained using Focal + Triplet loss with 5-fold stratified cross-validation, and applied PCA + isotonic-calibrated RBF-SVM for reliable predictions; achieved 99.8% test accuracy, 99.85% F1-score, and 99.97% ROC-AUC, with Grad-CAM, SHAP, and t-SNE for explainability.

Real-Time Multimodal Voice Agent with Agentic RAG & Self-Corrective Tool Orchestration
Python · FastAPI · LangGraph · WebRTC · faster-whisper · Silero VAD · Multimodal LLM · Qdrant · Redis · MCP · LoRA/QLoRA

- Built a real-time multimodal voice agent using streaming ASR/VAD, LangGraph multi-agent orchestration, Agentic RAG, MCP-based tool calling, and streaming TTS for low-latency multi-turn task execution, with barge-in handling for natural interruption.
- Implemented **self-corrective RAG**, hybrid BM25 + dense retrieval, reranking, claim-level verification, adaptive model routing, and **LoRA/QLoRA fine-tuning** to improve groundedness while reducing hallucination, latency, and unnecessary tool calls.
- Built an evaluation/observability framework measuring WER, task success, hallucination rate, retrieval quality, tool-call efficiency, TTFA, P95 latency, token usage, and cost.

Heart Disease Prediction System
Python · scikit-learn · XGBoost · Flask

- Built a predictive ML pipeline using Random Forest and XGBoost for heart disease classification; achieved 92% accuracy and 90% F1-score; deployed as a Flask web app for real-time patient data entry and risk prediction.

ACHIEVEMENTS & CERTIFICATIONS

- GATE CSE 2024 — Qualified (Graduate Aptitude Test in Engineering, Computer Science)